

MSE603_Assignment 1

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Q1:

1. (35 points) Use the simplex method to solve the following linear program:

$$\begin{aligned} \min \quad & z = x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 = 3 \\ & x_1 - 2x_2 \geq -7 \\ & x_1 + 5x_2 \leq 16 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Solution:

① First, introduce the variable z , and re-write the objective function $\min z$.

s.t.

$$\begin{aligned} z - x_1 - 2x_2 - Ma_1 &= 0 & \text{Row 0} \\ x_1 + 2x_2 + a_1 &= 3 & \text{Row 1} \\ -x_1 + 2x_2 + e_2 &= 7 & \text{Row 2} \\ x_1 + 5x_2 + s_3 &= 16 & \text{Row 3} \\ x_1, x_2, a_1, e_2, s_3 &\geq 0 \end{aligned}$$

② Then, we form the initial simplex tableau.

	Z	x_1	x_2	a_1	e_2	s_3	RHS	ratio
Row 0	1	-1	-2	-M	0	0	0	
Row 1	0	1	2	1	0	0	3	
Row 2	0	-1	2	0	1	0	7	
Row 3	0	1	5	0	0	1	16	

③ Then, we eliminate the artificial variable a_1 from the objective (Z)

$$R_0 \leftarrow R_0 + MR_1$$

$$Z - x_1 - 2x_2 - Ma_1 = 0 \rightarrow Z + (M-1)x_1 + (2M-2)x_2 = 3M$$

$$M(x_1 + 2x_2 + a_1) = 3M$$

We transfer the simplex tableau.

	Z	x_1	x_2	a_1	e_2	s_3	RHS	ratio
Row 0	1	M-1	2M-2	0	0	0	3M	
Row 1	0	1	2	1	0	0	3	3/2
Row 2	0	-1	2	0	1	0	7	7/2
Row 3	0	1	5	0	0	1	16	16/5

BV = $\{a_1, e_2, s_3\}$
NBV = $\{x_1, x_2\}$

④ From Row 2, we can know: $x_2 = \frac{3-x_1-a_1}{2}$

Then, we transfer the simplex tableau.

	Z	x_1	x_2	a_1	e_2	s_3	RHS	ratio
Row 0	1	0	0	(M-1)	0	0	3	
Row 1	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	0	$\frac{3}{2}$	
Row 2	0	-2	0	-1	1	0	4	
Row 3	0	$-\frac{3}{2}$	0	2.5	0	1	8.5	

BV = $\{x_2, e_2, s_3\}$
NBV = $\{x_1, a_1\}$

⑤ No positive coefficients in Z-row $\rightarrow$ solution is optimal.

NBV, $x_1 = 0, a_1 = 0$. BV: $x_2 = \frac{3}{2}, e_2 = 4, s_3 = 8.5$

$$Z = x_1 + 2x_2 = 3$$

Q2:

2. (30 points) Write down the dual of the problem in Q1 and find its optimal solution using duality results. Do not apply the simplex method directly to the dual problem.

Solution:

① First, we transfer primal problem to dual problem

Primal Problem:

$$\min Z = x_1 + 2x_2 \Rightarrow$$

$$\begin{aligned} \text{s.t. } x_1 + 2x_2 &= 3 & y_1 \\ x_1 - 2x_2 &\geq -7 & y_2 \\ x_1 + 5x_2 &\leq 16 & y_3 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Dual Problem:

$$\max W = 3y_1 - 7y_2 + 16y_3$$

$$\begin{aligned} \text{s.t. } y_1 + y_2 + y_3 &\leq 1 & x_1 \\ 2y_1 - 2y_2 + 5y_3 &\leq 2 & x_2 \\ y_1 \text{ is u.r.s, } y_2 \geq 0, y_3 \geq 0 \end{aligned}$$

② Then, we use the final simplex tableau in Q1.

From the Row 0 of the optimal tableau, we can know:

$$y_1 = (\text{coefficient of } a_1 \text{ in Row 0}) + M$$

$$y_1 = -(M-1) + M = 1$$

$$y_2 = (\text{coefficient of } e_2 \text{ in Row 0})$$

$$y_2 = 0$$

$$y_3 = (\text{coefficient of } s_3 \text{ in Row 0})$$

$$y_3 = 0$$

Therefore, the optimal value of the dual problem is:

$$W = 3y_1 - 7y_2 + 16y_3 = 3 \times 1 + 0 + 0 = 3$$

Q3:

3. (35 points) Consider the following LP:

$$\begin{aligned} \max \quad & z = 3x_1 + x_2 \\ \text{s.t.} \quad & 4x_1 + x_2 \leq 7 \\ & 5x_1 + 2x_2 \leq 10 \\ & x_1, x_2 \geq 0 \end{aligned}$$

The optimal solution to this LP is $z = \frac{17}{3}$, $x_1 = \frac{4}{3}$, $x_2 = \frac{5}{3}$. Use sensitivity analysis to determine the range of values for the right-hand side of both constraints (one at a time) for which the current basis remains optimal.

Solution:

① First, we express the linear program in matrix form.

The given LP can be written as: $\max Z = c^T X$, s.t. $AX = b$, $x \geq 0$

$$\max z = 3x_1 + x_2$$

$$\text{s.t. } 4x_1 + x_2 + s_1 = 7 \\ 5x_1 + 2x_2 + s_2 = 10$$

$$\rightarrow A = \begin{bmatrix} 4 & 1 & 10 \\ 5 & 2 & 0 & 1 \end{bmatrix}, b = \begin{bmatrix} 7 \\ 10 \end{bmatrix}, c = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

② Then, we identify the BN and NBN: $B = \begin{bmatrix} 4 & 1 \\ 5 & 2 \end{bmatrix}$, $N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

The inverse of B is:

$$B^{-1} = \frac{1}{4 \times 2 - 5 \times 1} \begin{bmatrix} 2 & -1 \\ -5 & 4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -5 & 4 \end{bmatrix} \quad (\text{formula: } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix})$$

③ Next, we consider the first case: only the first constraint changes.

The new RHS is: $b' = \begin{bmatrix} \bar{b}_1 \\ 10 \end{bmatrix}$. To keep the current basis optimal, the new basic solution must remain feasible: $x'B = B^{-1}b' \geq 0$

$$B^{-1}b' = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -5 & 4 \end{bmatrix} \begin{bmatrix} \bar{b}_1 \\ 10 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2\bar{b}_1 - 10 \\ -5\bar{b}_1 + 40 \end{bmatrix} \geq 0 \quad (\text{formula: } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax+by \\ cx+dy \end{bmatrix})$$

$$\begin{cases} 2\bar{b}_1 - 10 \geq 0 \\ -5\bar{b}_1 + 40 \geq 0 \end{cases} \rightarrow \begin{cases} \bar{b}_1 \geq 5 \\ \bar{b}_1 \leq 8 \end{cases} \quad \boxed{5 \leq b_1 \leq 8}$$

④ Similarly, we consider the second case; only the second constraint changes.

The new RHS is: $b' = \begin{bmatrix} 7 \\ \bar{b}_2 \end{bmatrix}$

$$B^{-1}b' = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -5 & 4 \end{bmatrix} \begin{bmatrix} 7 \\ \bar{b}_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 14 - \bar{b}_2 \\ -35 + 4\bar{b}_2 \end{bmatrix} \geq 0$$

$$\begin{cases} 14 - \bar{b}_2 \geq 0 \\ -35 + 4\bar{b}_2 \geq 0 \end{cases} \rightarrow \begin{cases} \bar{b}_2 \leq 14 \\ \bar{b}_2 \geq \frac{35}{4} \end{cases} \quad \boxed{\frac{35}{4} \leq b_2 \leq 14}$$

⑤ Conclusion: $5 \leq b_1 \leq 8$, $\frac{35}{4} \leq b_2 \leq 14$.